

Amazon Music Clustering

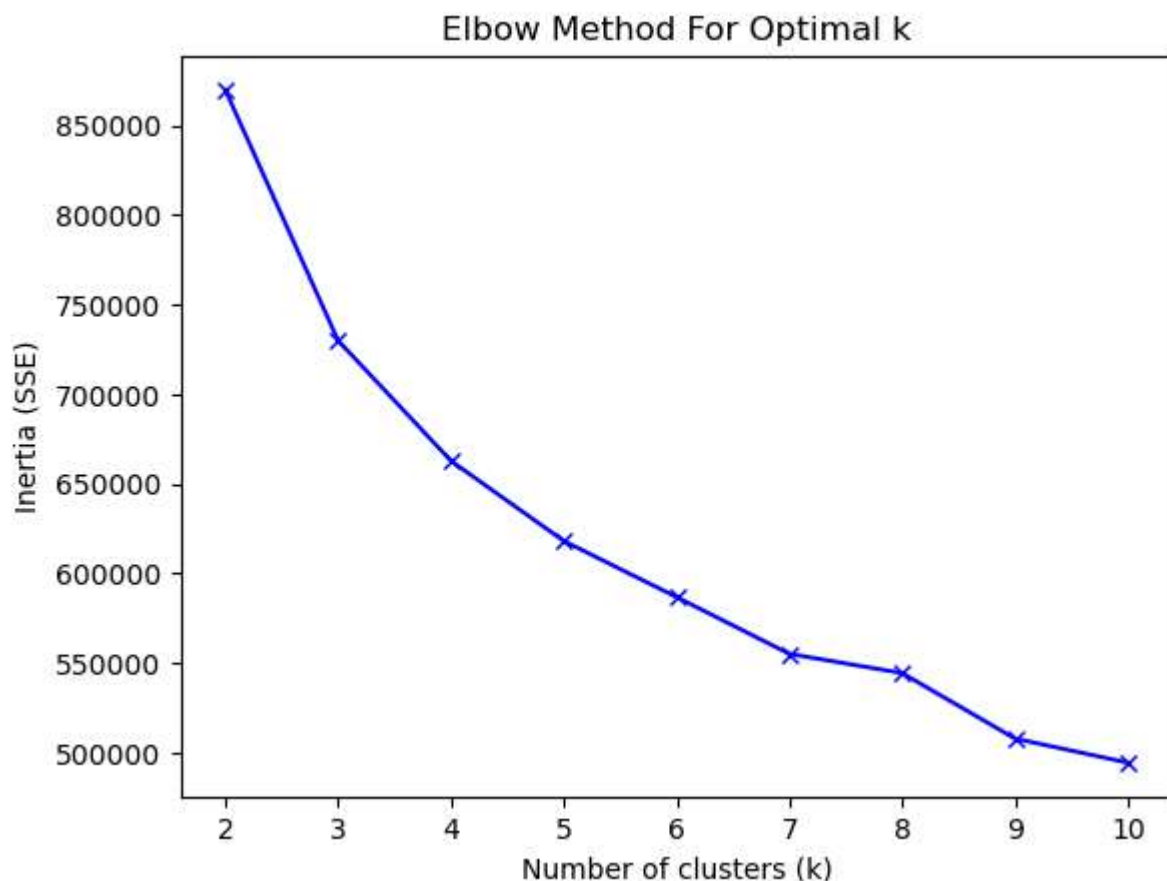
Interpretation: Clustering Techniques

Option: A - K-Means Clustering:

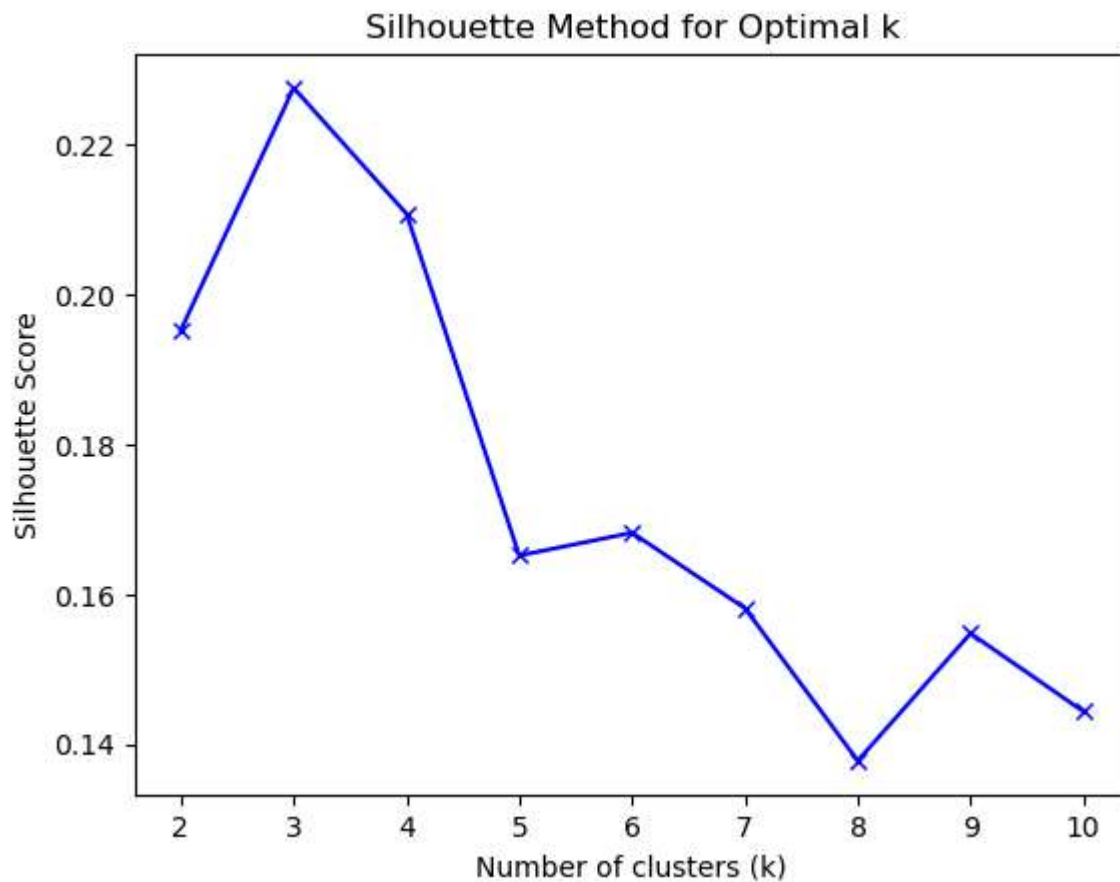
Goal: Apply one or more clustering algorithms to group the songs.

Step1:

- i) Find the best optimal K using Elbow Method
- ii) By assigning the $K = \text{range}(2,11)$
- iii) Looping those K values and by inertia

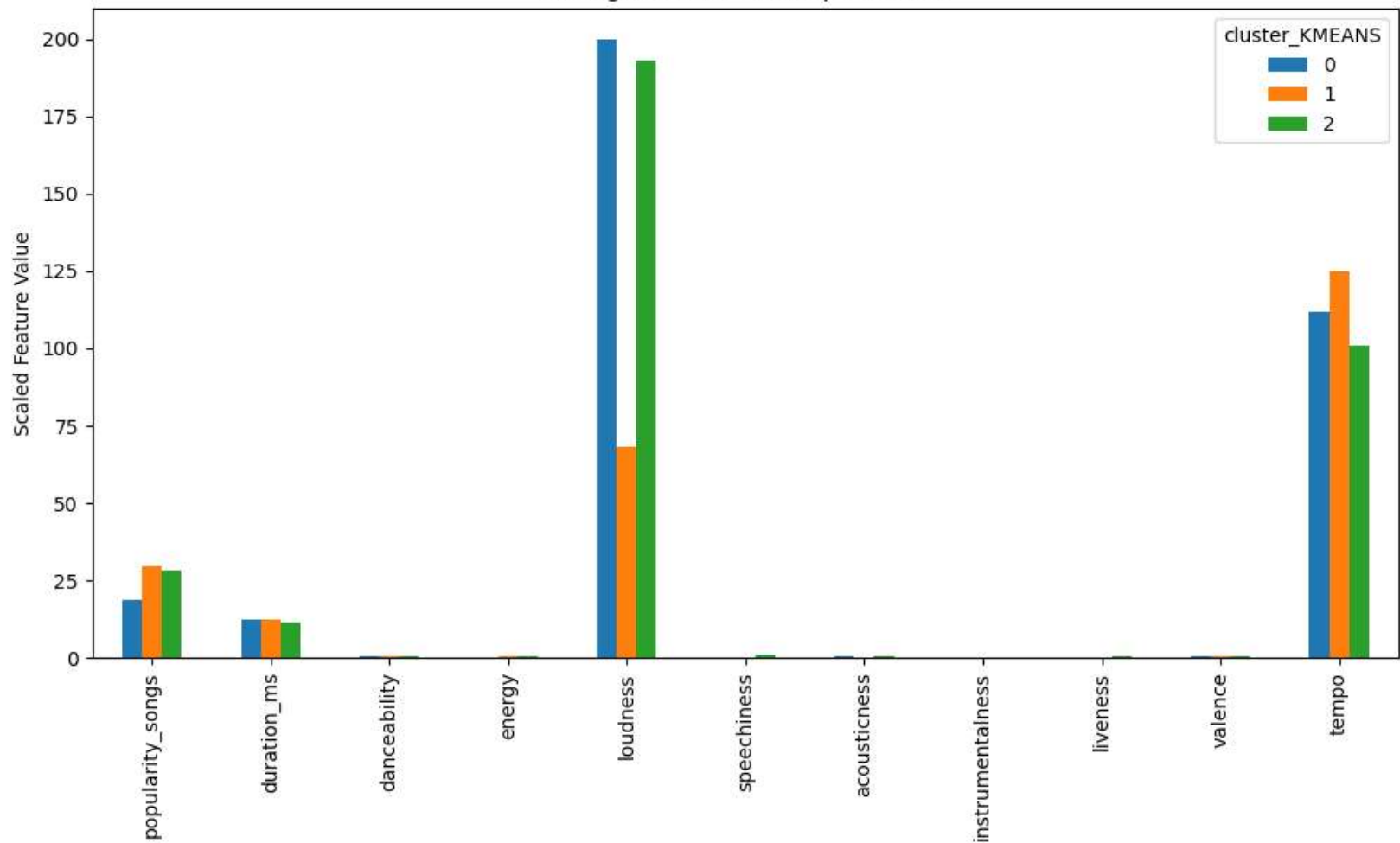


- iv) From the above Trend can able to find the optimal $K = 3$ but we had another step weather these K values perform well in silhouette_score evaluation.
- v) So in Evaluate process we are the same $K = \text{range}(2,11)$
- vi) By this can able to find which K value perform well

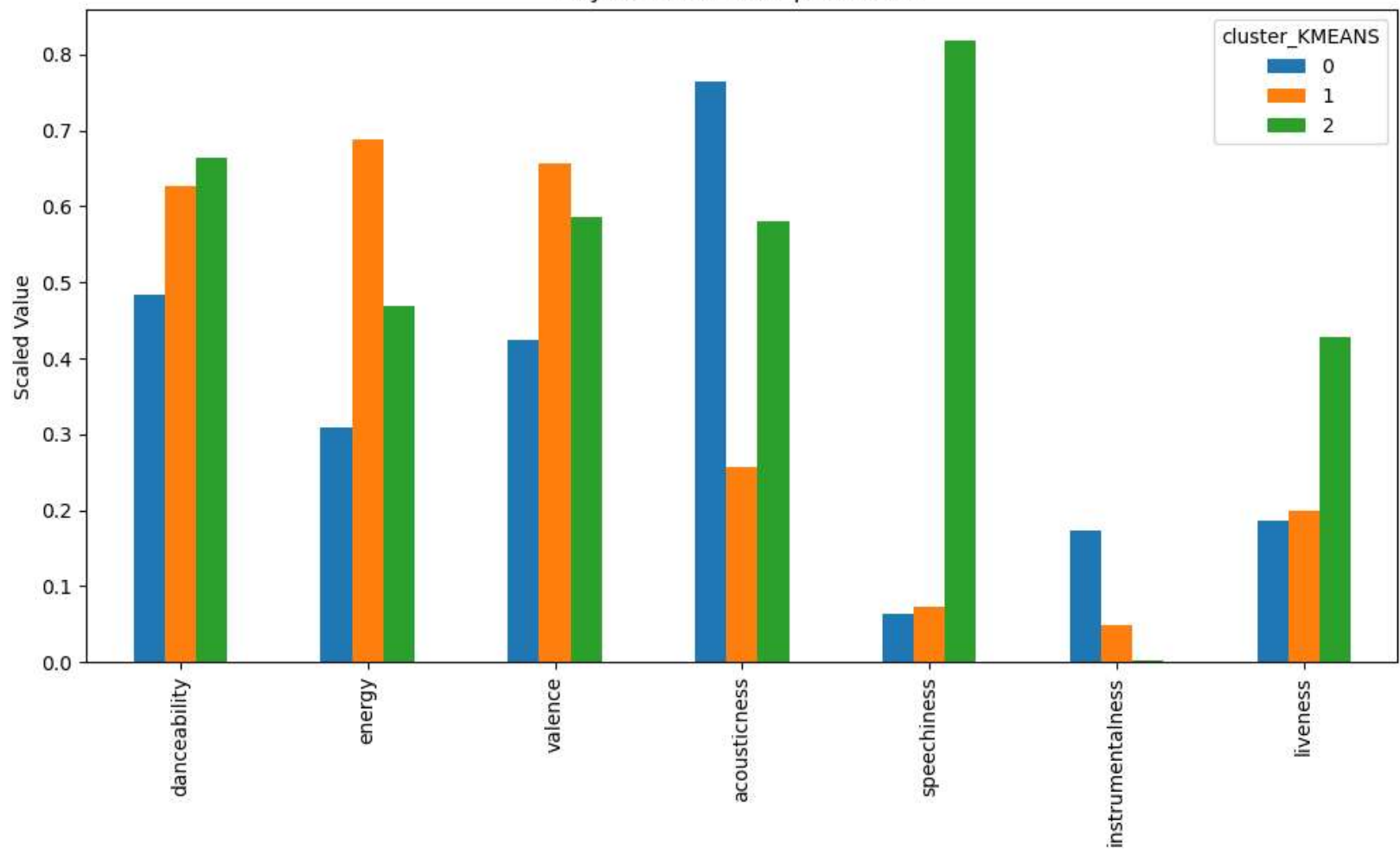


- vii) By the above trend can able to derive that best silhouette_score for K value 3 provides us a good score compare to others.
- viii) So we are approaching the clustering problem with $K = 3$.
- ix) After proceeding with $K = 3$ value we are assigning `df['cluster'] = kmeans.labels_`
- x) Then summarizing and groupby cluster with their `mean()`.

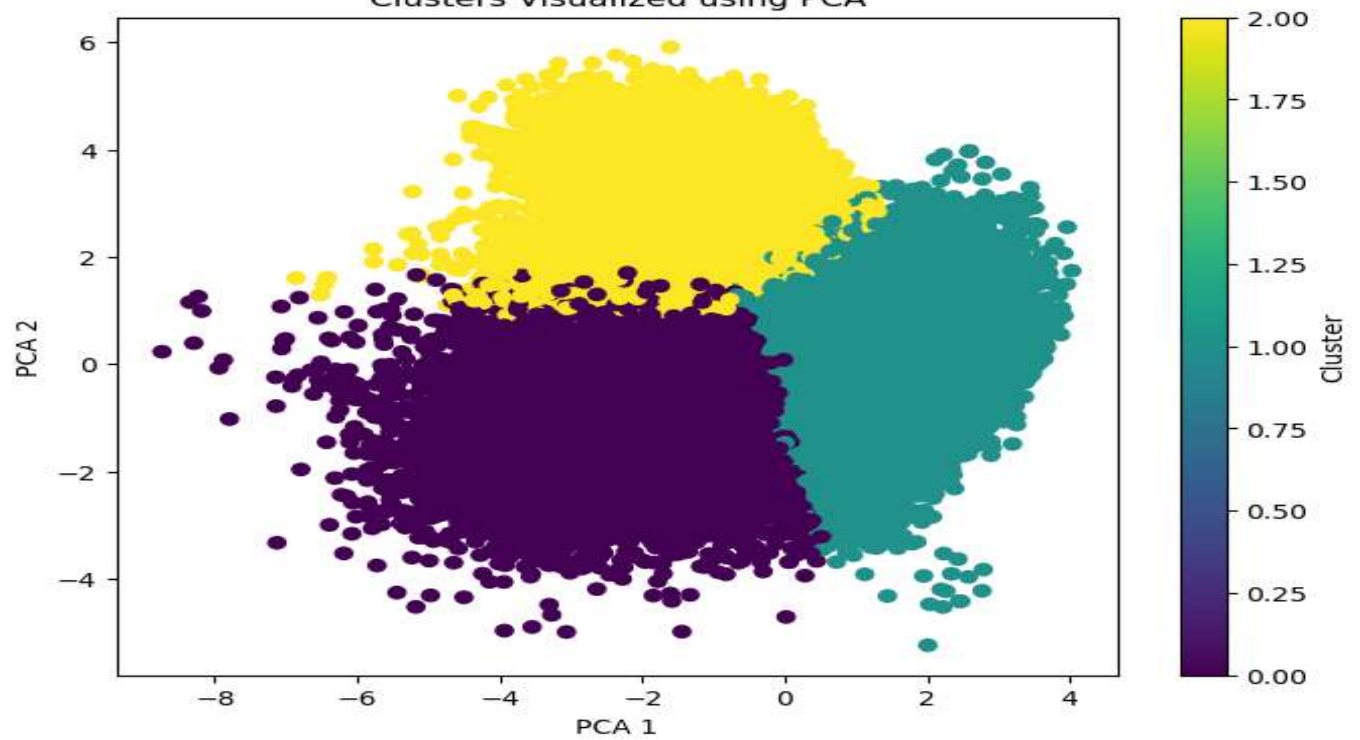
Average Feature Values per Cluster



Key Audio Features per Cluster



Clusters Visualized using PCA



Interpretation of Average Feature Values per Cluster:

The two charts together illustrate how the three clusters differ across both overall and key audio features.

Cluster 0 (Blue)

shows high loudness and acousticness **but** lower energy, danceability, and valence, **indicating songs that are** calm, acoustic, and less upbeat. **These tracks are likely to be** soft, relaxing, or unplugged versions **with lower tempo and energy.**

Cluster 1 (Orange)

stands out with **high energy, danceability, and valence**, along with **high tempo and moderate loudness.**

This suggests that songs in this cluster are **lively, energetic, and positive**, representing **party or feel-good tracks** with strong beats and higher rhythm intensity.

Cluster 2 (Green)

Has **balanced danceability and energy, higher speechiness and liveness**, and **slightly lower loudness and tempo** than the other two.

This profile indicates **live performances, vocal-heavy, or spoken-word style songs**, which are more expressive and interactive.

Overall Summary

- **Cluster 0:** Calm/Acoustic Tracks
- **Cluster 1:** Upbeat/Party Songs
- **Cluster 2:** Live/Vocal Tracks